Mohamed Azarudeen Allabaksh

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Education

Master of Science in Computer Science — Aug 2024 – May 2026
Stony Brook University, SUNY — Stony Brook, NY

Technical Skills

Languages: C#, Java, C++, Python, JavaScript, TypeScript, SQL, Shell/Bash
Frameworks & APIs: .NET Core, ASP.NET Core, Semantic Kernel, Spring Boot, gRPC, REST APIs, React, Angular
Azure: Azure Functions, AKS, Azure OpenAI, Event Hub, CosmosDB, Azure Monitor, LogAnalytics, Azure AI Content Safety
Cloud & DevOps: AWS (EC2, Lambda, S3, SQS), Kubernetes, Docker, GitHub Actions, Jenkins, Terraform, Prometheus, Grafana
AI/ML & Data: PyTorch, TensorFlow, scikit-learn, PySpark, Apache Spark, Pandas, NumPy, RAG Pipelines, LLM Agents, Vector Retrieval, Azure OpenAI
Distributed Systems: Apache Kafka, Azure Event Hub, Kafka Streams, High Availability, Fault-Tolerant Design

Experience

Software Engineer I — Aug 2024 – Jan 2026
Stony Brook University — New York, NY

- Architected large-scale data ingestion pipelines in **C# and .NET Core** on **Azure Functions and AKS**, processing 8K–12K structured entity records daily with **99.4% uptime** and sub-160ms p95 latency across 4 services.
- Built **Apache Spark** batch and micro-batch jobs in **PySpark** to aggregate and transform 3TB+ of structured entity data, feeding downstream **scikit-learn** and **PyTorch** scoring models that improved feature signal quality by **36%**.
- Built telemetry-driven observability using **Azure Monitor, LogAnalytics, Prometheus, and Grafana**, enabling proactive health analysis and reducing MTTR by **34%**.
- Automated **CI/CD pipelines** via **GitHub Actions and Terraform** with zero-downtime deployments, reducing release cycle from **24 min to 8 min** across 11 backend services.

Software Engineer Intern — May 2025 – Aug 2025
Lumina View — New York, NY

- Developed Responsible AI–aligned document intelligence using **C#, .NET Core, Azure OpenAI, and Semantic Kernel**, integrating LLM inference routing and orchestration to improve user productivity by **50%**.
- Designed retrieval and contextual ranking pipelines with **Azure AI Content Safety** guardrails and policy enforcement, reducing AI-generated content risk incidents by **41%** across 95K+ monthly inference requests.
- Implemented LLM evaluation infrastructure using **Azure Monitor** and **Pandas**-driven statistical validation, maintaining **99.3% safe response rate** and model reliability across production workflows.
- Integrated **ASP.NET Core** backend services via high-performance IPC, improving runtime stability by **23%** across 9 production deployments.

Software Engineer I — Dec 2023 – Aug 2024
Mavdero Techservices — Chennai, India

- Engineered high-availability distributed services in **C# and Java** handling **95K+ daily transactions** with optimized concurrency and p95 latency under **140ms**.
- Built scalable **ASP.NET Core** REST and gRPC microservices integrated with **Apache Kafka**, improving backend throughput by **17%** across 5 production services.
- Delivered **PySpark** and **Apache Spark** batch jobs for structured data transformation across 14 pipeline stages, cutting reporting latency from **4.2s to 0.9s** and reducing **Pandas**-based preprocessing time by **43%**.
- Trained and deployed **scikit-learn** and **TensorFlow** models for transaction anomaly detection, achieving **91.4%** precision and reducing false-positive alert volume by **28%** in production.

Projects

AI-Powered Entity Enrichment & Ranking Platform — *C#, .NET Core, Azure Functions, CosmosDB, Azure OpenAI, PySpark, PyTorch, scikit-learn*

- Designed a large-scale entity enrichment service using **C#, Azure Functions, CosmosDB, and Azure OpenAI**, processing **500K+ location records** through **PySpark** map-reduce jobs and **Azure Event Hub** ingestion pipelines.
- Trained **PyTorch** and **scikit-learn** models on **NumPy/Pandas**-processed feature sets to score and rank entity signals, improving data quality by **29%** against a 2M-record staging dataset.
- Implemented red-teaming validation with **Azure Monitor** dashboards and **Pandas** anomaly detection, ensuring Responsible AI compliance and **99.2% safe output rate** across inference workflows.

Responsible AI Fraud Detection Platform — *C#, Azure Functions, Event Hub, CosmosDB, Azure OpenAI, Semantic Kernel, TensorFlow*

- Architected a Responsible AI–aligned fraud detection service using **C#, Azure Functions, Event Hub, and CosmosDB**, processing **95K+ daily transactions** and reducing revenue leakage by **26%**.
- Built a **TensorFlow**-backed scoring layer with **Pandas** and **NumPy** feature pipelines, delivering **Semantic Kernel**–orchestrated LLM inference across **8 production rule sets**.
- Built monitoring pipelines using **Azure Monitor and LogAnalytics**, tracking model reliability across 12K–15K daily inference calls with p95 latency under **180ms**.